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Database Systems

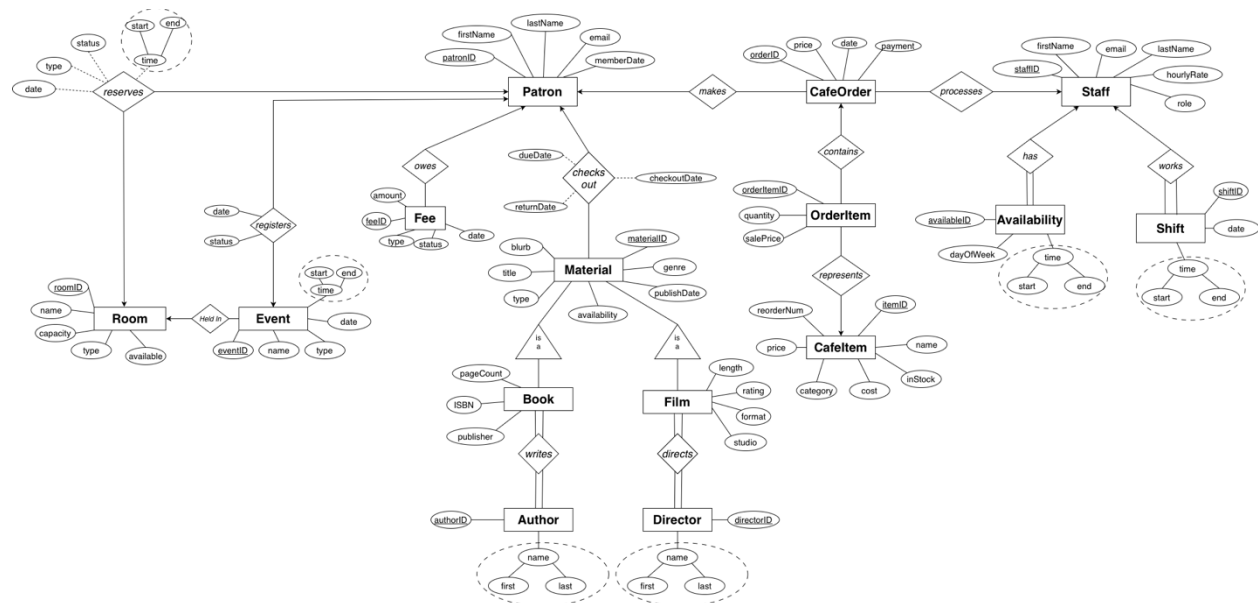
1 May 2026

Project Deliverable 5: Database Implementation

1 Project Description

This project is a database management system for a library. The purpose of the system is to help organize and manage important library operations such as material management, room reservations, event registrations, patron fees, cafe sales, and cafe inventory. Many different types of people can use it. For example, a patron can use it to view their checkout materials, room reservations, event registrations, unpaid fees, past café orders, and to search through all the library materials. A staff member can use the website to create new patron accounts, view patron information, view checkouts, view café sales, update patron fees, and delete patron accounts. The website uses a variety of SQL tools such as queries, functions, procedures, and views to show all of this information to the users.

2 ER Model



3 Data Dictionary

<u>Data Item</u>	<u>Definition</u>	<u>Data Type</u>	<u>Notes</u>
authorID	Unique integer ID for each author.	Numeric	Cannot be Null Primary Key
availability	A string indicating if the material is available or checked out	String	
blurb	A one sentence preview/summary/description of the book or film.	String	
checkoutDate	YYYY-MM-DD formatted date for when a material was checked out	Date?	
directorID	Unique integer ID for each director.	Numeric	Cannot be Null Primary Key
dueDate	YYYY-MM-DD formatted date for when a material must be returned.	Date?	
email	A string representing the patron's email address.	String	
eventID	A unique integer number that represents the different events.	Int	Not Null Primary Key
eventType	A string representing what kind of event is being held.	String	“Class”/ “Group”/ “Academic”/etc.
feeAmount	A floating point number representing the dollar and cent amount of the fee.	Float	Must be above 0
feeID	A unique integer number for each fee in the system.	Int	Not Null Primary Key
feeStatus	A string stating if the fee has been paid or not.	String	“Paid” or “Unpaid”
feeType	A string representing the reason for the fee.	String	“Late”/“Damaged”/etc.
filmFormat	A string representing the storage format of the film (DVD, VHS, etc.)	String	
filmLength	Runtime in minutes of a film.	Numeric	

firstName	A string representing the user's first name.	String	
genre	A string representing the literary or film genre of the material.	String	
ISBN	A 13 digit number that each book receives (shared by books that are multiple copies of the same book)	Numeric	
lastName	A string representing the user's last name.	String	
materialID	Unique integer ID for each material in the collection.	Numeric	Cannot be Null Primary Key
materialType	A string representing if the material is a book or a film	String	
memberDate	YYYY-MM-DD formatted date for when a patron made a library account.	Date?	
name	A string representing a user's first and last name formatted FIRST LAST that combines firstName and lastName	String	
pageCount	A number representing the number of pages in a book	Numeric	
patronID	Unique integer ID for each patron who has a library account.	Numeric	Cannot be Null Primary Key
publishDate	YYYY-MM-DD formatted date for when a material in the collection was published or released	Date?	
publisher	A string representing the company or organization that published the book.	String	
rating	A string indicating the rating for age-recommendation to watch the film	String	
registrationStatus	A string representing if the patron has been accepted for the event.	String	“accepted”, “denied”, or “pending”
reservationStatus	A string representing if the reservation was accepted for the room.	String	“accepted”, “denied”, or “pending”

reservationType	A string representing why the patron is reserving the room.	String	“Study”/ “Group” / “Lab”/ etc.
returnDate	YYYY-MM-DD formatted date for when a material in the collection was returned	Date?	
roomAvailable	Represents whether the room is available during a certain time.	Bool	
roomCapacity	An integer number representing the amount of people that the room can hold.	Int	Must be greater than 0
roomID	A unique integer number representing each room the library has to offer to patrons.	Int	Not Null Primary Key
roomType	A string representing the type of room.	String	“Class” / “Lab” / “Media” / etc.
studio	A string representing the company or organization that managed or financed the film	String	
title	A string representing the name or title of the material in the collection	String	
orderID	A unique integer number representing each order placed	Int	Not Null AutoIncremented? Primary Key
orderPrice	The amount a cafe order costs	Int	Greater Than or equal to 0
orderDate	YYYY-MM-DD formatted date for when a cafe order was placed	Date?	
orderPayment	A predefined string value representing if an order has been paid for or not	String	“paid” / “unpaid”
orderItemID	A unique integer number representing each type of item in a cafe order	Int	Not Null Primary Key
quantity	An integer number tracking how much of one item there is in an order	Int	Must be greater than or equal to 0
salePrice	An number that show how much the sale price is for the item/items in an order	Int	Must be greater than or equal to 0
itemID	A unique integer number representing a different item that the cafe sells	Int	Not Null Primary Key
itemName	A string representing the name of an item in the cafe	String	Not Null

inStock	An integer number that represents how much of an item the cafe has currently in stock	Int	Must be greater than or 0 Not Null
itemCost	A number that shows how much the item cost the cafe to buy it	Int	Must be greater than or equal to 0 Not Null
category	A predefined string that represents the category a cafe item belongs to	String	“Hot Drink”/ “Cold Drink”/ “Pastry”/ “Sandwich” / “Snack” / “Fruit”
itemPrice	A number that represents how much an item will cost before tax for the customer	Int	Must be greater than or equal to 0 Not Null
reorderNum	An integer number that represents how often an item is ordered at the cafe	Int	Must be greater than or equal to 0
staffID	A unique integer number representing each staff member who works at the library	Int	Not Null AutoIncrement? Primary Key
hourlyRate	A number that represents the hourly rate of a staff member	Int	Must be greater than or equal to 0
role	A predefined string value that represents the staff member’s position/role at the library	String	“Manager” / “Barista” / “Cashier” / “Supervisor” / “Librarian”
ShiftID	A unique integer representing the different shifts available to work	Int	Not Null Primary Key
availableID	A unique integer representing the different available times of a staff member	Int	Not Null Primary Key